Tutorial - 6

$(\ker T, (\ker T)^\perp)$

Q1: Let $T: \mathbb{F}^n \to \mathbb{F}^n$ be a linear map. Show that

$$\det(T(\alpha_1), \ldots, T(\alpha_n)) = c \det(\alpha_1, \alpha_2, \ldots, \alpha_n)$$

for all $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{F}^n$, where

$$c = \det(T(e_1), \ldots, T(e_n)).$$ and

$e_1, e_2, \ldots e_n$ is the standard basis for $\mathbb{F}^n$.

Sol^n Let $x \in \mathbb{F}^n$

We can write x in terms of standard basis of $\mathbb{F}^n$

Let $x_i = \sum_{j=1}^{n} a_{ij} e_j$ for some $a_{ij} \in \mathbb{F}$

Since $T: \mathbb{F}^n \to \mathbb{F}^n$

$$T(x_i) = T\left(\sum_{j=1}^{n} a_{ij} e_j\right) = \sum_{j=1}^{n} a_{ij} T(e_j)$$

Thus $T(x_i)$ is a linear combination of transformed vectors $T(e_1), T(e_2) \ldots T(e_n)$.

We can express the transformed vectors $T(x_1), T(x_2), \ldots T(x_n)$ as the columns of a matrix where each column is a linear ~~column~~ combination of $T(e_1), T(e_2) \ldots T(e_n)$.

In matrix form,

$$\text{Matrix}[T(x_1), T(x_2) \ldots, T(x_n)] = [T(e_1), T(e_2), \ldots T(e_n)] \cdot \text{Matrix}[a_{ij}]$$

$$\det(T(x_1), T(x_2), \cdots T(x_n)) = \det[T(e_1), T(e_2), \cdots T(e_n)] \cdot \det(x_1, x_2, \cdots x_n)$$

$$\Rightarrow \det(T(x_1), T(x_2), \cdots T(x_n)) = c \det(x_1, x_2, \cdots x_n)$$

Q2: Find matrices $A, B \in M_{n\times n}(\mathbb{C})$ and scalars $\lambda, \beta \in \mathbb{C}$ such that

(i) λ is an eigenvalue of A, β is an eigenvalue of B but $\lambda+\beta$ is not an eigenvalue of $A+B$.

(ii) $\lambda\beta$ is not an eigenvalue of AB.

Sol^n Let, $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ for $n=2$

(i) Eigen values of $A = 2, 3$ Let $\lambda = 2$

" " $B = 1, 2$ Let $\beta = 2$

~~But~~ $[A+B] = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$ eigenvalue of $[A+B] = 3, 5$

Clearly $\lambda+\beta = 4$ is not an eigenvalue of $[A+B]$

(ii) $[AB] = \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix}$.

Clearly $\lambda\beta = 4$ is not e.v. of $[AB]$

Q3: Suppose λ is an eigenvalue of $A \in M_{n\times n}(\mathbb{F})$. Show that λ^n is an eigenvalue of A^n.

Moregenerally, $p(\lambda)$ is an eigen value of $p(A)$, for any polynomial p.

Sol^n: Since λ is e.v. of A $\exists$ $v \neq 0$ s.t. ~~$A\lambda = \lambda$~~ $Av = \lambda v$ — (i)

Now, Consider, $A^n v = (A A \cdots A) v$ (n times)

$$= (A A \cdots A)(Av)$$
$$= (A A \cdots A)(\lambda v) \quad \text{by (i)}$$
$$= \lambda (AA \cdots A) v \quad (n-1) \text{ times}$$
$$= \lambda (AA \cdots A)(Av)$$
$$= \lambda (AA \cdots A)(\lambda v) \quad \text{by (i)}$$
$$= \lambda^2 (A A \cdots A) v$$
$$\vdots$$
$$= \lambda^n v$$

hence $\underline{A^n v = \lambda^n v}$ i.e. λ^n is an e.v. of A^n.

Now, Let p be any pol.

then $p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_k x^k$

Thus for matrix A,

$$P(A) = a_0 + a_1 A + a_2 A^2 + \cdots + a_k A^k$$

Now Consider $p(A)v = (a_0 + a_1 A + a_2 A^2 + \cdots + a_k A^k) v$

$$P(A)\, v = a_0 v + a_1 A \cdot v + a_2 A^2 \cdot v + \cdots + a_k A^k \cdot v$$
$$= a_0 v + a_1 \lambda v + a_2 \lambda^2 \cdot v + \cdots + a_k \lambda^k \cdot v \quad \text{(by part(i))}$$
$$= (a_0 + a_1 \lambda + a_2 \lambda^2 + \cdots + a_k \lambda^k)\, v$$
$$= P(\lambda) v$$

Q 4: Let $A \in M_{n\times n}(\mathbb{C})$ be a Nilpotent matrix i.e. $A^r = 0$ for some $r \in \mathbb{N}$. Find the eigenvalues of A.

Sol^n: Let λ be an eigenvalue of $A \in M_{n\times n}(\mathbb{C})$

then $\exists$ vector $v \neq 0$ s.t. $\quad Av = \lambda v$

Now, $\quad$ Since $\quad A^r v = \lambda^r v \quad$ (by previous ques)

$$0 = \lambda^r . v$$

v is not zero so it is only possible when $\lambda = 0$

Since λ was arbitary hence All e.v. of A are zeros. ◪

Q5: Show that Similar matrices has same eigenvalues.

Sol^n: Let A and B are similar matrices. then $\exists$ matrix (invertible) s.t. $\quad B = P^{-1}AP$. — $(*)$

Now let λ is eigen value of A then $\exists$ $v \neq 0$ s.t. $Av = \lambda v$ —(i)

To show: λ is e.v. of B.

Since $\quad Av = \lambda v$

$$P^{-1}Av = P^{-1}(\lambda v)$$

$$\cancel{P^{-1}A = P^{-1}\lambda \quad \text{—(ii)}}$$

Now, Consider, $\quad B(PV) = P^{-1}AP(PV) = P^{-1}\lambda\, P(PV)$

$$= \lambda P^{-1}P(PV) \quad \text{by (ii)}$$

$$= \lambda(PV)$$

hence λ is e.v. of B w.r.t. eigenvector (PV) ◪

Consider, $$AV = \lambda V$$

$$P^{-1}AV = P^{-1}\lambda V$$

$$P^{-1}AV = \lambda(P^{-1}V)$$

$$P^{-1}APP^{-1}V = \lambda(P^{-1}V)$$

$$B(P^{-1}V) = \lambda(P^{-1}V)$$

hence λ is eigenvalue of B w.r.t. eigenvector $P^{-1}V$

B